

Panel Data Econometrics

Chapter 3 – The GLS estimator *Random-effect model*

Reminder

In the previous chapter, we presented the error component model.

We discussed three estimators: the OLS (pooling model), the Within estimator and the Between estimator.

The OLS estimator can be expressed as a (weighted) combination of the two other estimators.

The OLS estimator is not efficient, because its variance contains Ω . We can still use it, as long as we correct the variance-covariance matrix of the estimator.

Reminder

The Ω matrix

For the full sample, the variance-covariance matrix (size NT) of the error is a block diagonal matrix given by :

$$\Omega = I_N \otimes (\sigma_v^2 I_T + \sigma_\eta^2 J_T) = \sigma_v^2 I_{NT} + \sigma_\eta^2 (I_N \otimes J_T)$$

Introduction

In this third chapter, we discuss ...

Part 1 – The GLS estimator

Part 2 – The FGLS estimators

Part 3 – Applications

The GLS estimator

The Generalized Least Squares estimator

The Generalized Least Squares estimator considers the individual effects as random draws from a specific distribution, and seeks to estimate the parameters of this distribution:

→ We are not interested in quantifying the different individual effects.

This estimator is often referred to as the

« Random-Effect model »

The GLS estimator

The Generalized Least Squares estimator

Note : It is very important that we still assume that covariates are not correlated with the errors (exogeneity).

The GLS estimator is written :

$$\hat{\boldsymbol{\beta}}^{GLS} = (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{y}$$

Its variance is

$$\text{var}(\hat{\boldsymbol{\beta}}^{GLS}) = (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}$$

The GLS estimator

The Generalized Least Squares estimator

It is possible to disentangle between the two sources of variations (*intra-individual* and *inter-individual*).

$$\hat{\beta}^{GLS} = \left(\frac{1}{\sigma_v^2} X' W X + \frac{1}{\sigma_l^2} X' B X \right)^{-1} \left(\frac{1}{\sigma_v^2} X' W y + \frac{1}{\sigma_l^2} X' B y \right)$$

$$var(\hat{\beta}^{GLS}) = \left(\frac{1}{\sigma_v^2} X' W X + \frac{1}{\sigma_l^2} X' B X \right)^{-1}$$

The GLS estimator

The Generalized Least Squares estimator

$$\hat{\beta}^{GLS} = \left(\frac{1}{\sigma_v^2} X' W X + \frac{1}{\sigma_l^2} X' B X \right)^{-1} \left(\frac{1}{\sigma_v^2} X' W y + \frac{1}{\sigma_l^2} X' B y \right)$$

If the variance of the individual effect (σ_η^2) **dominates**, this estimator will approach the **Within estimator**.

If the variance of the individual effect (σ_η^2) **is dominated**, this estimator will approach the **OLS estimator**.

The GLS estimator

The Generalized Least Squares estimator

To estimate the GLS estimator, **one must know the content of the variance-covariance matrix Ω .**

- Premultiply all variables (X and y) by $\Omega^{-\frac{1}{2}}$

$$X^* = \Omega^{-\frac{1}{2}}X ; y^* = \Omega^{-\frac{1}{2}}y$$

- Estimate the model, with the transformed variables, using OLS.

$$\hat{\beta}^{GLS} = (X^{*'}X^*)^{-1}X^{*'}y^*$$

The GLS estimator

Summary

The OLS estimator is not efficient if $\sigma_{\eta}^2 \neq 0$, as our standard assumption A4 is not respected anymore.

The GLS estimator offers an efficient estimator of β . To use it correctly however, one must know Ω , the variance-covariance matrix of the errors.

The GLS estimator will always provide values that are between the Within estimator ($\sim$ when individual effects are strong) and the OLS estimator ($\sim$ when individual effects are negligible).

The GLS estimator

Récapitulatif en français

L'estimateur des MCO n'est pas efficace si $\sigma_{\eta}^2 \neq 0$, puisque notre hypothèse standard A4 n'est plus respectée.

L'estimateur des MCG constitue un estimateur efficace de β . Pour l'utiliser correctement, cependant, il faut avoir connaissance de Ω , la matrice de variance-covariance des termes d'erreur.

L'estimateur des MCG prendra toujours des valeurs situées entre l'estimateur Within ($\sim$ quand les effets individuels sont forts) et l'estimateur des MCO ($\sim$ quand les effets individuels sont négligeables).

The FGLS estimator

The Feasible Generalized Least Squares estimator

In our journey toward the efficient estimator, we meet a significant issue:

We do not know Ω !

Reminder: Ω is a block diagonal matrix, where the diagonal contains $\Omega_1, \Omega_2, \dots, \Omega_N$

$$\text{With } \Omega_i = \begin{pmatrix} \sigma_v^2 + \sigma_\eta & \sigma_\eta & \dots & \sigma_\eta \\ \sigma_\eta & \sigma_v^2 + \sigma_\eta & \dots & \sigma_\eta \\ \dots & \dots & \dots & \dots \\ \sigma_\eta & \sigma_\eta & \dots & \sigma_v^2 + \sigma_\eta \end{pmatrix}$$

The FGLS estimator

The Feasible Generalized Least Squares estimator

Our goal will be to estimate $\mathbf{\Omega}$ with $\widehat{\mathbf{\Omega}}$. To do that, we need to estimate $\widehat{\sigma}_l^2$ and $\widehat{\sigma}_v^2$.

Natural estimators would be

$$\widehat{\sigma}_l^2 = \frac{\mathbf{\varepsilon}' \mathbf{B} \mathbf{\varepsilon}}{N}$$

$$\widehat{\sigma}_v^2 = \frac{\sum_i \sum_t (\varepsilon_{it} - \bar{\varepsilon}_{i.})^2}{N(T-1)}$$

We do not observe the error terms ε , but we may be able to use the residuals (if they come from a consistent estimation of beta !)

The FGLS estimator

The Feasible Generalized Least Squares estimator

The most straightforward correction would be to use residuals from the OLS estimators. This is the Wallace and Hussain (1969) estimator.

Wallace, T. D., & Hussain, A. (1969). The use of error components models in combining cross section with time series data. *Econometrica: Journal of the Econometric Society*, 55-72.

$$\widehat{\sigma}_l^2 = \frac{e'_{OLS} B e_{OLS}}{N}$$

$$\widehat{\sigma}_v^2 = \frac{e'_{OLS} W e_{OLS}}{N(T - 1)}$$

$$\widehat{\sigma}_\eta^2 = \frac{\widehat{\sigma}_l^2 - \widehat{\sigma}_v^2}{T}$$

The FGLS estimator

The Feasible Generalized Least Squares estimator

The Amemiya (1971) estimator is based on residuals issued from the Within estimator.

Amemiya, T. (1971). The estimation of the variances in a variance-components model. *International Economic Review*, 1-13.

We first compute the overall intercept

$$\hat{\beta}_0 = \bar{\bar{y}} - \hat{\beta}^W X$$

We then compute the residuals

$$e_w = y - \hat{\beta}_0 - \hat{\beta}^W X$$

Problem : the variance of the individual effect is overestimated if the model contains some time-invariant variables which disappear with the within transformation.

The FGLS estimator

The Feasible Generalized Least Squares estimator

The Hausman and Taylor (1981) estimator corrects the issue discussed in the previous slide.

Hausman, J. A., & Taylor, W. E. (1981). Panel data and unobservable individual effects. *Econometrica: Journal of the Econometric society*, 1377-1398.

e_{HT} are calculated as the residuals from the regression of e_W on all time-invariant variables dropped in the process of estimating the Within estimator. This reduces the estimation of $\hat{\sigma}_l^2$ (therefore $\hat{\sigma}_\eta^2$) who could overestimate the variance of the individual effects.

The FGLS estimator

The Feasible Generalized Least Squares estimator

The Swamy and Arora (1972) estimator uses residuals from both the Within and the Between estimators.

Swamy, P. A. V. B., & Arora, S. S. (1972). The exact finite sample properties of the estimators of coefficients in the error components regression models. *Econometrica: journal of the Econometric Society*, 261-275.

$$\hat{\sigma}_l^2 = \frac{e_B' B e_B}{N - K - 1}$$

$$\hat{\sigma}_v^2 = \frac{e_W' W e_W}{N(T - 1) - K}$$

$$\hat{\sigma}_\eta^2 = \frac{\widehat{\sigma}_l^2 - \widehat{\sigma}_v^2}{T}$$

The FGLS estimator

The Feasible Generalized Least Squares estimator

All previous methods estimate σ_η^2 by subtracting σ_v^2 from σ_l^2 . In contrast, Nerlove (1971) estimates directly σ_η^2 from the Within model, as the empirical variance of the fixed effects.

Nerlove, M. (1971). A note on error components models. *Econometrica: Journal of the Econometric Society*, 383-396.

$$\begin{aligned}\hat{\eta}_i &= \bar{y}_{i.} - \hat{\beta}^W \bar{x}_{i.} \\ \hat{\sigma}_\eta^2 &= \sum_i \frac{(\hat{\eta}_i - \bar{\hat{\eta}})^2}{N-1} \\ \hat{\sigma}_v^2 &= \frac{e_W' W e_W}{NT}\end{aligned}$$

The FGLS estimator

Summary

We do not know Ω but several methods allow us to estimate $\hat{\Omega}$.

All those estimators of Ω generally give very similar results. There are then used to estimate the FGLS estimator (applying OLS to $\hat{\Omega}^{-\frac{1}{2}}X$ and $\hat{\Omega}^{-\frac{1}{2}}y$).

The random-effect model is a FGLS estimator. It means that:

- It is biased, but consistent (if A3 is respected).
- It is more precise than the OLS estimator if there are indeed individual effects in our panel data model.

The FGLS estimator

Récapitulatif (en français)

Nous ne connaissons pas Ω mais plusieurs méthodes permettent d'estimer $\hat{\Omega}$.

Tous ces estimateurs de Ω offre généralement des résultats assez proches. Ils sont utilisés pour calculé l'estimateur des MCQG (appliquer les MCO à $\hat{\Omega}^{-\frac{1}{2}}X$ et $\hat{\Omega}^{-\frac{1}{2}}y$).

Le modèle à effets aléatoires est un estimateur des Moindres Carrés Quasi-Généralisés. Cela signifie que:

- Cet estimateur est biaisé, mais convergent (si A3 est respectée).
- Cet estimateur est plus précis que celui des MCO, dès lors qu'il y a effectivement des effets individuels dans notre modèle sur données de panel.

Application

The feasible generalized least squares estimator

Application with TobinQ

We get the data from the pder package. It includes, among other variables :

cusip : compustat's identifying number

year : year

ikn : investment divided by capital : narrow definition

qn : Tobin's Q : narrow definition

- See R code –

Application

The feasible generalized least squares estimator

Application with ForeignTrade

Kinal, T., & Lahiri, K. (1993). On the estimation of simultaneous-equations error-components models with an application to a model of developing country foreign trade. *Journal of Applied Econometrics*, 8(1), 81-92.

We estimate the relationship between imports and Gross National Product (real, per capita)

- See R Code -

Application

The feasible generalized least squares estimator

Application with TurkishBank

El-Gamal, M. A., & Inanoglu, H. (2005). Inefficiency and heterogeneity in Turkish banking: 1990–2000. *Journal of Applied Econometrics*, 20(5), 641-664.

We estimate the cost function of banks.

- See R Code -

Application

The feasible generalized least squares estimator

Application with TexasElectr

Kumbhakar, S. C. (1996). Estimation of cost efficiency with heteroscedasticity: an application to electricity utilities. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 45(3), 319-335.

We estimate the cost function of electric production in Texas.

- See R Code -